

1. Even or Odd [EvenorOdd_309.cpp]

```
#include <iostream>
using namespace std;

int main() {
    int num;
    if (cin >> num) {
        if (num % 2 == 0)
            cout << num << " is Even\n";
        else
            cout << num << " is Odd\n";
    }
    return 0;
}
```

2. Find Maximum of Two Numbers [FindMaximum_309.cpp]

```
#include <iostream>
using namespace std;

int main() {
    int a, b;
    if (cin >> a >> b) {
        cout << (a > b ? a : b) << "\n";
    }
    return 0;
}
```

3. Sum of First N Natural Numbers

```
#include <iostream>
using namespace std;

int main() {
    int n;
    if (cin >> n) {
        int sum = 0;
        for (int i = 1; i <= n; ++i) {
            sum += i;
        }
        cout << sum << "\n";
    }
    return 0;
}
```

4. Check if a Number is Prime

```
#include <iostream>
using namespace std;

int main() {
    int num;
    if (cin >> num) {
        if (num <= 1) {
            cout << "Not Prime\n";
            return 0;
        }
        bool isPrime = true;
        for (int i = 2; i * i <= num; ++i) {
            if (num % i == 0) {
                isPrime = false;
                break;
            }
        }
        if (isPrime)
            cout << "Prime\n";
        else
            cout << "Not Prime\n";
    }
    return 0;
}
```

5. Factorial of a Number

```
#include <iostream>
using namespace std;

int main() {
    int n;
    if (cin >> n) {
        long long fact = 1;
        for (int i = 1; i <= n; ++i) {
            fact *= i;
        }
        cout << fact << "\n";
    }
    return 0;
}
```

6. Reverse a Number

```
#include <iostream>
using namespace std;

int main() {
    int num;
    if (cin >> num) {
        int rev = 0;
        while (num != 0) {
            rev = rev * 10 + (num % 10);
            num /= 10;
        }
        cout << rev << "\n";
    }
    return 0;
}
```

7. Check if a Number is Palindrome

```
#include <iostream>
using namespace std;

int main() {
    int num;
    if (cin >> num) {
        int original = num, rev = 0;
        while (num > 0) {
            rev = rev * 10 + (num % 10);
            num /= 10;
        }
        if (original == rev)
            cout << "Palindrome\n";
        else
            cout << "Not Palindrome\n";
    }
    return 0;
}
```

8. Print First N Fibonacci Numbers

```
#include <iostream>
using namespace std;

int main() {
    int n;
    if (cin >> n) {
        long long a = 0, b = 1;
        for (int i = 0; i < n; ++i) {
            cout << a << (i == n - 1 ? "" : " ");
            long long next = a + b;
            a = b;
            b = next;
        }
        cout << "\n";
    }
    return 0;
}
```

9. Count Digits of a Number

```
#include <iostream>
using namespace std;

int main() {
    int num;
    if (cin >> num) {
        int temp = num < 0 ? -num : num;
        int count = (temp == 0) ? 1 : 0;
        while (temp > 0) {
            count++;
            temp /= 10;
        }
        cout << count << " digits\n";
    }
    return 0;
}
```

10. Sum of Digits of a Number

```
#include <iostream>
using namespace std;

int main() {
    int num;
    if (cin >> num) {
        int temp = num < 0 ? -num : num;
        int sum = 0;
        while (temp > 0) {
            sum += temp % 10;
            temp /= 10;
        }
        cout << sum << "\n";
    }
    return 0;
}
```

11. Linear Search

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    int n;
    if (cin >> n) {
        vector<int> arr(n);
        for (int i = 0; i < n; ++i) cin >> arr[i];
        int key;
        cin >> key;
        int index = -1;
        for (int i = 0; i < n; ++i) {
            if (arr[i] == key) {
                index = i;
                break;
            }
        }
        if (index != -1)
            cout << "Found at index " << index << "\n";
        else
            cout << "Not found\n";
    }
    return 0;
}
```

12. Swap Two Numbers

```
#include <iostream>
using namespace std;

int main() {
    int a, b;
    if (cin >> a >> b) {
        int temp = a;
        a = b;
        b = temp;
        cout << "a = " << a << ", b = " << b << "\n";
    }
    return 0;
}
```

13. Simple Calculator

```
#include <iostream>
using namespace std;

int main() {
    double num1, num2;
    char op;
    if (cin >> num1 >> op >> num2) {
        switch (op) {
            case '+': cout << "Result: " << num1 + num2 << "\n"; break;
            case '-': cout << "Result: " << num1 - num2 << "\n"; break;
            case '*': cout << "Result: " << num1 * num2 << "\n"; break;
            case '/':
                if (num2 != 0) cout << "Result: " << num1 / num2 << "\n";
                else cout << "error: division by zero\n";
                break;
            default: cout << "Invalid operator\n";
        }
    }
    return 0;
}
```

14. Print Even Numbers from 1 to N

```
#include <iostream>
using namespace std;

int main() {
    int n;
    if (cin >> n) {
        for (int i = 2; i <= n; i += 2) {
            cout << i << (i + 2 > n ? "" : " ");
        }
        cout << "\n";
    }
    return 0;
}
```

15. Find Second Largest Number

```
#include <iostream>
#include <vector>
#include <climits>
using namespace std;

int main() {
    int n;
    if (cin >> n) {
        vector<int> arr(n);
        for (int i = 0; i < n; ++i) cin >> arr[i];

        int first = INT_MIN, second = INT_MIN;
        for (int val : arr) {
            if (val > first) {
                second = first;
                first = val;
            } else if (val > second && val < first) {
                second = val;
            }
        }
        if (second != INT_MIN)
            cout << "Second largest: " << second << "\n";
        else
            cout << "no second largest found\n";
    }
    return 0;
}
```

16. Find Average of Array Elements

```
#include <iostream>
#include <vector>
#include <iomanip>
using namespace std;

int main() {
    int n;
    if (cin >> n) {
        vector<double> arr(n);
        double sum = 0;
        for (int i = 0; i < n; ++i) {
            cin >> arr[i];
            sum += arr[i];
        }
        cout << fixed << setprecision(1) << "Average: " << sum / n << "\n";
    }
    return 0;
}
```

17. Remove Duplicates from Array

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    int n;
    if (cin >> n) {
        vector<int> arr(n);
        for (int i = 0; i < n; ++i) cin >> arr[i];

        vector<int> uniqueArr;
        for (int i = 0; i < n; ++i) {
            bool duplicate = false;
            for (int j = 0; j < i; ++j) {
                if (arr[i] == arr[j]) {
                    duplicate = true;
                    break;
                }
            }
            if (!duplicate) uniqueArr.push_back(arr[i]);
        }

        for (size_t i = 0; i < uniqueArr.size(); ++i) {
            cout << uniqueArr[i] << (i + 1 == uniqueArr.size() ? "" : " ");
        }
        cout << "\n";
    }
    return 0;
}
```

18. Convert Lowercase to Uppercase

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string str;
    if (getline(cin, str)) {
        for (char &ch : str) {
            if (ch >= 'a' && ch <= 'z') {
                ch = ch - 32;
            }
        }
        cout << str << "\n";
    }
    return 0;
}
```